INCUBATOR MONITORING SYSTEM

Sushma A, Asst. professor
Dept. of Information Science & Engineering
Nagarjuna College of Engineering & Technology,
Bengaluru, Karnataka, India
sushma.ash20@gmail.com

Harshitha N, Student
Dept. of Information Science & Engineering
Nagarjuna College of Engineering & Technology,
Bengaluru, Karnataka, India
Harshitha.nchari@gmail.com

Vandana K M, Student
Dept. of Information Science & Engineering
Nagarjuna College of Engineering & Technology,
Bengaluru, Karnataka, India
vandanamgowda3@gmail.com

Pothu Sai Visal Reddy
Dept. of Information Science & Engineering
Nagarjuna College of Engineering & Technology,
Bengaluru, Karnataka, India
vishal3827@gmail.com

ABSTRACT - The neonatal incubator is an apparatus that provides a closed and managed environment for the sustenance of temperature babies. But recently, many premature babies have lost their lives due to lack of appropriate monitoring of the incubator that leads to accidents (leakage of gas and overheating causing life threat and eventually, the bursting of incubators).This mission offers with the cost-tremendous plan of an embedded system that monitors certain parameters such as pulse charge of the baby, temperature, humidity, incubator. If any veriations happens in the corresponding parameters (pulse, temperature and humidity), than the parameters will manage by way of the usage of lights and followers inside the incubator. And the readings will consistently monitoring in LCD and sends the important points to the corresponding medical doctor or nurse by using the usage of GSM and the baby will monitors continuously by means of webpage. The parents, doctors or nurse will reveal the child from somewhere by the usage of IOT.

I. INTRODUCTION:

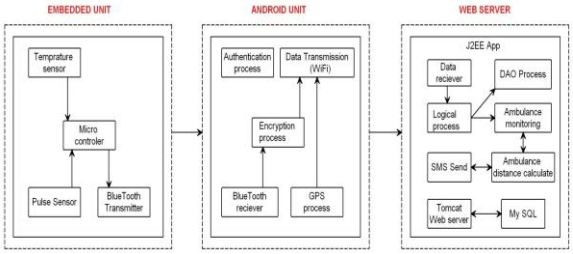
The increase in neonatal deaths from untimely birth and low birth-weight was found to be non-uniform across India such that the demise fees were extra in rural areas and poorer states however lesser in urban areas and in richer states. Moreover, premature births and low birth-weight children may also require extra investments in incubators and intensive care units in order to furnish fabulous neonatal care. There have additionally been incidents of deaths of untimely infants due to accidents which have been identified as a technical fault in the incubators. The current system proposes the use of a temperature sensor to sense the temperature of the incubator which is connected to the Node MCU. Any amplify in the temperature past the detailed vary turns the alarm on and the heater in the incubator receives grew to become off via the utilization of a cell app. This allows of workers(doctors, parents) to get notification at some stage in an emergency so that imperative preventive moves can be taken. This will ensures the protection of temperature internal the incubator. But still there are many other parameters that may effects infants in incubator which as to be monitored and managed to equip safety to the infant. The important goal of this project is to overcome the demise of infants and provide a safe and low-cost technology for monitoring the incubator which will help in reduce the mortality rate of neonates. The proposed device includes the use of WIFI, integrated to various sensor units, such as pulse, temperature and humidity sensors.

II. EXISTING SYSYTEM:

Many of the incubator structures contains the temperature and other parameters displayed on the screen, which is fixed to the incubator system. In the maternity ward, the doctors or nurses should be monitoring health condition of infants once in every hour or vice versa. The presence of a doctor or a nurse is always required and they need to continuously monitor the health status of an infant in the incubator. This will not be accurate in taking care all the time, sometimes due to human negligence demise can happen. And in the present incubator systems accumulated statistics from the sensor cannot transfer to the far away locations and the incubator device can't manipulate data from the remote locations.

III. PROPOSED SYSTEM:

In incubator device we offers a homogeneous and steady temperature, relative humidity stage and it constantly monitors the infant heart beat level and additionally alert the difficulty individual with an alert message and offers live toddler monitoring. The parents additionally screen the toddler stay condition. The assignment “Incubator infant parameters controlling and monitoring” is a very progressive machine which will assist to keep the kiddies in heat condition. This gadget monitors the untimely baby by way of webpage. And ship all parameters data to the mobile through the use of GSM.



Figuare: SYSTEM ARCHITECTURE

The system will be managed routinely via sensors and microcontroller chips placed inside the incubator. The incubator would possibly be started in the heating mode to heat up till reached to a saturation point, then switched off the temperature regulator automatically, in case the temperature raised over critical point ie. 98.6 F (37.0 C), the fan will be switched on to modulate it to the room temperature air interior the incubator until it hits the wished degree. In addition, heart rate measurements device nearby to in order to monitoring hemodynamic status. We are using Wi-Fi module as the wi-fi conversation medium and as controller we are using ARDUINO UNO. First we will measure the temperature and humidity of the incubator and reveal the gadget the use of Wi-Fi module to webpage. If the data is above/below a set factor then the controller will switching ON/OFF the fan and light. The readings will be shown on the LCD screen. The LM35, dht11and Pulse sensors will provide non-stop statistics and so we can recognize the temperature and humidity and heart-beat at any time. So using this model we can effortlessly control the temperature, humidity and measure the infant’s heartbeat. Two and the readings will send to mobile the usage of GSM. By the use of digicam which is linked to raspberry pi, will supply stay monitoring of the infant in webpage.

IV. SYSTEM REQUIREMENTS

(i) Temperature sensor-LM35

LM35 device is a temperature sensor device. It is connected to Arduino. The device senses variations in temperature inside incubator. LM35 is a basic temperature sensor, which is used for experimental purpose. It analyzes the readings in centigrade (degree Celsius) since its output voltage is directly proportional to temperature. If temperature increases, the voltage across diode increases at known rate (the voltage drop occurs across base-emitter junction of transistor).

(ii) Pulse sensor –M212

It consists of a LED light that is deployed to measure the pulse rate. Based on the volume of blood in the capillaries, the light gets reflected as the sensor is placed on the body. So, the amount of reflection taking place during heartbeat will be less than that with no heartbeat. Also the volume of blood inside the capillaries decreases in between heart beats, which affects the transmission of light through the tissues This variation in transmission and reflection of light gives the analog pulse output from the sensor. Working The system consists of Raspberry pi which have inbuilt wi-fi. It is connected to the incubator. The system has various sensors like temperature, humidity and pulse are connected to Arduino and they will control by using controller. The controller will read and collects the data continuously and it will sends the information to the corresponding doctor using GSM. The temperature and humidity sensor senses the temperature of the surroundings and the humidity present in the surrounding environment of the neonate. The pulse sensor also continuously takes the readings of baby. If the temperature and humidity values exceed the specified range (36.5-37.2°C) or when the pulse rate varies, then the respective Sensors will senses the values and controller will reads the values from sensors and controls the temperature and humidity by turn on/off the light and fan. The controller will sends the message to concern person the system will provides live monitoring of baby via webpage from anywhere so, the parent or doctor will continuously monitor the baby actions in live monitoring.

V. RESULT

The project “Incubator baby parameter sensing and monitoring” was designed by using sensors and IOT technology. And the status of the baby will be monitored by using webpage.

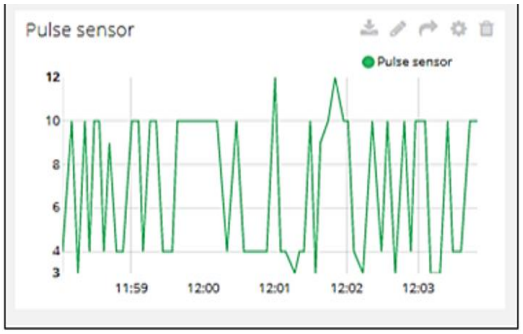


Figure: PULSE RATE SENSOR



Figure: MESSAGE ON WEB PAGE

VI. CONCLUSION

Mutual authentication is essential for wireless medical data transmission between the users in the infant incubator monitoring system. Also, the system needs to provide user authentication and data integrity by using limited resources. But, the conventional mutual authentication protocols are not suitable for resource limited devices. Therefore, in this paper, a computationally efficient and effective, new encryption scheme based mutual authentication protocol is presented to improve the accessing or computational efficiency with balanced information security by using constraint resources and devices in the system. The security analysis and performance analysis of the proposed computationally efficient mutual authentication protocol ensure that the proposed system outperforms when compared.

Finally, incubator monitoring and fetching neonatal critical condition to the authenticated person can outbreak infant mortality.

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